

Date : 22/sep/2025

Team : skyops crew

Topic : node js deployment on EC2

1. Launch a EC2 instance and connect it with server.

The screenshot displays the AWS Management Console interface for the 'Instances' page. The top navigation bar shows the AWS logo, a search bar, and the current region 'United States (N. Virginia)'. The left sidebar contains the 'EC2' menu with options like Dashboard, EC2 Global View, Events, Instances, Instance Types, Launch Templates, Spot Requests, Savings Plans, Reserved Instances, Dedicated Hosts, Capacity Reservations, Images, AMIs, AMI Catalog, Elastic Block Store, and Volumes. The main content area shows a table of instances with one instance named 'nodejs' (ID: i-003f13bdb22fac9e0) in a 'Running' state. Below the table, the 'Details' tab for the instance 'i-003f13bdb22fac9e0 (nodejs)' is selected, showing fields for Instance ID, Public IPv4 address (54.91.61.15), Private IPv4 addresses (172.31.30.133), Instance state (Running), and Public DNS (ec2-54-91-61-15.compute-1.amazonaws.com).

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv
nodejs	i-003f13bdb22fac9e0	Running	t3.micro	Initializing	View alarms +	us-east-1a	ec2-54-91

Instance summary		
Instance ID	Public IPv4 address	
i-003f13bdb22fac9e0	54.91.61.15 open address	
IPv6 address	Instance state	Private IPv4 addresses
-	Running	172.31.30.133
		Public DNS
		ec2-54-91-61-15.compute-1.amazonaws.com open

Command :-

```
, #_  
~\_ #####_ Amazon Linux 2023  
~~ \#####\  
~~ \###|  
~~ \#/ __ https://aws.amazon.com/linux/amazon-linux-2023  
~~ V~' '->  
~~~ /  
~~._. _/  
_/_/  
_/_/m/'
```

[ec2-user@ip-172-31-30-133 ~]\$ sudo su -

[root@ip-172-31-30-133 ~]# git clone
https://github.com/CloudTechDevOps/sample_nodejs_on_Ec2.git

-bash: git: command not found

[root@ip-172-31-30-133 ~]# sudo dnf install -y nodejs

Amazon Linux 2023 Kernel Livepatch repository
kB/s | 23 kB 00:00

178

Dependencies resolved.

=====			
=====			
==			
Package	Architecture	Version	Repository
Size			
=====			
=====			
==			

Installing:

nodejs	x86_64	1:18.20.8-1.amzn2023.0.2
amazonlinux	13 M	

Installing dependencies:

libbrotli	x86_64	1.0.9-4.amzn2023.0.2	amazonlinux
315 k			

nodejs-libs	x86_64	1:18.20.8-1.amzn2023.0.2
amazonlinux	14 M	

Installing weak dependencies:

nodejs-docs	noarch	1:18.20.8-1.amzn2023.0.2
amazonlinux	7.8 M	

nodejs-full-i18n	x86_64	1:18.20.8-1.amzn2023.0.2
amazonlinux	8.4 M	

nodejs-npm	x86_64	1:10.8.2-1.18.20.8.1.amzn2023.0.2
amazonlinux	1.9 M	

Transaction Summary

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Install 6 Packages

Total download size: 45 M

Installed size: 224 M

Downloading Packages:

(1/6): libbrotli-1.0.9-4.amzn2023.0.2.x86_64.rpm	5.8
MB/s 315 kB	00:00

(2/6): nodejs-docs-18.20.8-1.amzn2023.0.2.noarch.rpm	
59 MB/s 7.8 MB	00:00

(3/6): nodejs-full-i18n-18.20.8-1.amzn2023.0.2.x86_64.rpm	
52 MB/s 8.4 MB	00:00

(4/6): nodejs-18.20.8-1.amzn2023.0.2.x86_64.rpm	
46 MB/s 13 MB	00:00

(5/6): nodejs-npm-10.8.2-1.18.20.8.1.amzn2023.0.2.x86_64.rpm	
23 MB/s 1.9 MB	00:00

(6/6): nodejs-libs-18.20.8-1.amzn2023.0.2.x86_64.rpm	
56 MB/s 14 MB	00:00

Total	110 MB/s 45 MB
00:00	

Running transaction check

Transaction check succeeded.

Running transaction test

Transaction test succeeded.

Running transaction

Preparing :

1/1

Installing : nodejs-docs-1:18.20.8-1.amzn2023.0.2.noarch
1/6

Installing : libbrotli-1.0.9-4.amzn2023.0.2.x86_64
2/6

Installing : nodejs-libs-1:18.20.8-1.amzn2023.0.2.x86_64
3/6

Installing : nodejs-full-i18n-1:18.20.8-1.amzn2023.0.2.x86_64
4/6

Installing : nodejs-npm-1:10.8.2-1.18.20.8.1.amzn2023.0.2.x86_64
5/6

Installing : nodejs-1:18.20.8-1.amzn2023.0.2.x86_64
6/6

Running scriptlet: nodejs-1:18.20.8-1.amzn2023.0.2.x86_64
6/6

INFO: registered node-18 in the alternatives

Verifying : libbrotli-1.0.9-4.amzn2023.0.2.x86_64
1/6

Verifying : nodejs-1:18.20.8-1.amzn2023.0.2.x86_64
2/6

Verifying : nodejs-docs-1:18.20.8-1.amzn2023.0.2.noarch
3/6

Verifying : nodejs-full-i18n-1:18.20.8-1.amzn2023.0.2.x86_64
4/6

Verifying : nodejs-libs-1:18.20.8-1.amzn2023.0.2.x86_64
5/6

Verifying : nodejs-npm-1:10.8.2-1.18.20.8.1.amzn2023.0.2.x86_64
6/6

Installed:

libbrotli-1.0.9-4.amzn2023.0.2.x86_64 nodejs-1:18.20.8-1.amzn2023.0.2.x86_64
nodejs-docs-1:18.20.8-1.amzn2023.0.2.noarch

nodejs-full-i18n-1:18.20.8-1.amzn2023.0.2.x86_64 nodejs-libs-1:18.20.8-
1.amzn2023.0.2.x86_64 nodejs-npm-1:10.8.2-1.18.20.8.1.amzn2023.0.2.x86_64

Complete!

[root@ip-172-31-30-133 ~]# yum install git -y

Last metadata expiration check: 0:00:16 ago on Mon Sep 22 17:34:16 2025.

Dependencies resolved.

=====

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Package Size	Architecture	Version	Repository
=====			
=====			
==			

Installing:

git 53 k	x86_64	2.50.1-1.amzn2023.0.1	amazonlinux
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Installing dependencies:

git-core 4.9 M	x86_64	2.50.1-1.amzn2023.0.1	amazonlinux
git-core-doc amazonlinux	noarch 2.8 M	2.50.1-1.amzn2023.0.1	
perl-Error amazonlinux	noarch 41 k	1:0.17029-5.amzn2023.0.2	
perl-File-Find amazonlinux	noarch 25 k	1.37-477.amzn2023.0.7	
perl-Git 41 k	noarch	2.50.1-1.amzn2023.0.1	amazonlinux
perl-TermReadKey amazonlinux	x86_64 36 k	2.38-9.amzn2023.0.2	
perl-lib 15 k	x86_64	0.65-477.amzn2023.0.7	amazonlinux

Transaction Summary

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Install 8 Packages

Total download size: 7.9 M

Installed size: 41 M

Downloading Packages:

(1/8): git-2.50.1-1.amzn2023.0.1.x86_64.rpm 1.4
MB/s | 53 kB 00:00

(2/8): perl-Error-0.17029-5.amzn2023.0.2.noarch.rpm
2.0 MB/s | 41 kB 00:00

(3/8): git-core-doc-2.50.1-1.amzn2023.0.1.noarch.rpm
33 MB/s | 2.8 MB 00:00

(4/8): perl-File-Find-1.37-477.amzn2023.0.7.noarch.rpm
775 kB/s | 25 kB 00:00

(5/8): perl-Git-2.50.1-1.amzn2023.0.1.noarch.rpm 2.0
MB/s | 41 kB 00:00

(6/8): perl-TermReadKey-2.38-9.amzn2023.0.2.x86_64.rpm
1.5 MB/s | 36 kB 00:00

(7/8): git-core-2.50.1-1.amzn2023.0.1.x86_64.rpm
32 MB/s | 4.9 MB 00:00

(8/8): perl-lib-0.65-477.amzn2023.0.7.x86_64.rpm
329 kB/s | 15 kB 00:00

Total 44 MB/s | 7.9 MB
00:00

Running transaction check

Transaction check succeeded.

Running transaction test

Transaction test succeeded.

Running transaction

Preparing : 1/1

Installing : git-core-2.50.1-1.amzn2023.0.1.x86_64
1/8

Installing : git-core-doc-2.50.1-1.amzn2023.0.1.noarch
2/8

Installing : perl-lib-0.65-477.amzn2023.0.7.x86_64
3/8

Installing : perl-TermReadKey-2.38-9.amzn2023.0.2.x86_64
4/8

Installing : perl-File-Find-1.37-477.amzn2023.0.7.noarch
5/8

Installing : perl-Error-1:0.17029-5.amzn2023.0.2.noarch
6/8

Installing : perl-Git-2.50.1-1.amzn2023.0.1.noarch
7/8

Installing : git-2.50.1-1.amzn2023.0.1.x86_64
8/8

Running scriptlet: git-2.50.1-1.amzn2023.0.1.x86_64
8/8

Verifying : git-2.50.1-1.amzn2023.0.1.x86_64
1/8

Verifying : git-core-2.50.1-1.amzn2023.0.1.x86_64
2/8

Verifying : git-core-doc-2.50.1-1.amzn2023.0.1.noarch
3/8

Verifying : perl-Error-1:0.17029-5.amzn2023.0.2.noarch
4/8

Verifying : perl-File-Find-1.37-477.amzn2023.0.7.noarch
5/8

Verifying : perl-Git-2.50.1-1.amzn2023.0.1.noarch
6/8

Verifying : perl-TermReadKey-2.38-9.amzn2023.0.2.x86_64
7/8

Verifying : perl-lib-0.65-477.amzn2023.0.7.x86_64
8/8

Installed:

git-2.50.1-1.amzn2023.0.1.x86_64	git-core-2.50.1-1.amzn2023.0.1.x86_64	git-
core-doc-2.50.1-1.amzn2023.0.1.noarch		
perl-Error-1.0.17029-5.amzn2023.0.2.noarch	perl-File-Find-1.37-477.amzn2023.0.7.noarch	
perl-Git-2.50.1-1.amzn2023.0.1.noarch		
perl-TermReadKey-2.38-9.amzn2023.0.2.x86_64	perl-lib-0.65-477.amzn2023.0.7.x86_64	

Complete!

```
[root@ip-172-31-30-133 ~]# git --version
```

```
git: '--' is not a git command. See 'git --help'.
```

```
[root@ip-172-31-30-133 ~]# git clone
https://github.com/CloudTechDevOps/sample_nodejs_on_Ec2.git
```

```
Cloning into 'sample_nodejs_on_Ec2'...
```

```
remote: Enumerating objects: 45, done.
```

```
remote: Counting objects: 100% (14/14), done.
```

```
remote: Compressing objects: 100% (14/14), done.
```

```
remote: Total 45 (delta 7), reused 0 (delta 0), pack-reused 31 (from 1)
```

```
Receiving objects: 100% (45/45), 14.77 KiB | 14.77 MiB/s, done.
```

```
Resolving deltas: 100% (16/16), done.
```

```
[root@ip-172-31-30-133 ~]# cd nodejs-on-ec2
```

```
npm install
```

```
-bash: cd: nodejs-on-ec2: No such file or directory
```

```
npm error code ENOENT
```

```
npm error syscall open
```

```
npm error path /root/package.json
```

```
npm error errno -2
```

```
npm error enoent Could not read package.json: Error: ENOENT: no such file or directory, open
'/root/package.json'
```

```
npm error enoent This is related to npm not being able to find a file.
```

```
npm error enoent
```

```
npm error A complete log of this run can be found in: /root/.npm/_logs/2025-09-
22T17_35_37_572Z-debug-0.log
```

```
[root@ip-172-31-30-133 ~]# cd nodejs-on-ec2
```

```
-bash: cd: nodejs-on-ec2: No such file or directory
```



```
[root@ip-172-31-30-133 ~]# cd ^[[200~sample_nodejs_on_Ec2~
-bash: cd: $'\E[200~sample_nodejs_on_Ec2~': No such file or directory
[root@ip-172-31-30-133 ~]# cd sample_nodejs_on_Ec2
[root@ip-172-31-30-133 sample_nodejs_on_Ec2]# npm install

npm warn ancient lockfile
npm warn ancient lockfile The package-lock.json file was created with an old version of npm,
npm warn ancient lockfile so supplemental metadata must be fetched from the registry.
npm warn ancient lockfile
npm warn ancient lockfile This is a one-time fix-up, please be patient...
npm warn ancient lockfile

up to date, audited 1 package in 379ms

found 0 vulnerabilities
[root@ip-172-31-30-133 sample_nodejs_on_Ec2]# npm -v
10.8.2
[root@ip-172-31-30-133 sample_nodejs_on_Ec2]# npm install

up to date, audited 1 package in 372ms

found 0 vulnerabilities
[root@ip-172-31-30-133 sample_nodejs_on_Ec2]# npm start

> node-hello@1.0.0 start
> node index.js
```


PM2 is a Production Process Manager for Node.js applications
with a built-in Load Balancer.

Start and Daemonize any application:

```
$ pm2 start app.js
```

Load Balance 4 instances of api.js:

```
$ pm2 start api.js -i 4
```

Monitor in production:

```
$ pm2 monitor
```

Make pm2 auto-boot at server restart:

```
$ pm2 startup
```

To go further checkout:

<http://pm2.io/>

[PM2] Spawning PM2 daemon with pm2_home=/root/.pm2

[PM2] PM2 Successfully daemonized

[PM2] Starting /root/sample_nodejs_on_Ec2/index.js in fork_mode (1 instance)

[PM2] Done.

id	name	namespace	version	mode	pid	uptime	↻	status	cpu	mem
user		watching								

```
| 0 | node-app | default | 1.0.0 | fork | 26812 | 0s | 0 | online | 0% | 40.1mb |
root | disabled |
```

```
|-----|
|-----|
```

[root@ip-172-31-30-133 sample_nodejs_on_Ec2]# pm2 status node-app

```
|-----|
|-----|
```

```
| id | name   | namespace | version | mode | pid  | uptime | ↻ | status | cpu | mem
| user | watching |
```

```
|-----|
|-----|
```

```
| 0 | node-app | default | 1.0.0 | fork | 26812 | 45s | 0 | online | 0% | 50.5mb |
root | disabled |
```

```
|-----|
|-----|
```

[root@ip-172-31-30-133 sample_nodejs_on_Ec2]# pm2 stop index.js --name node-app

[PM2] Applying action stopProcessId on app [index.js](ids: [0])

[PM2] [node-app](0) ✓

```
|-----|
|-----|
```

```
| id | name   | namespace | version | mode | pid  | uptime | ↻ | status | cpu | mem
| user | watching |
```

```
|-----|
|-----|
```

```
| 0 | node-app | default | 1.0.0 | fork | 0    | 0    | 0 | stopped | 0% | 0b | root
| disabled |
```

```
|-----|
|-----|
```

[root@ip-172-31-30-133 sample_nodejs_on_Ec2]# pm2 status node-app

```
|-----|
|-----|
```

```
| id | name   | namespace | version | mode | pid  | uptime | ↻ | status | cpu | mem
| user | watching |
```

```
|-----|
|-----|
```

id	name	namespace	version	mode	pid	uptime	status	cpu	mem	user	watching
0	node-app	default	1.0.0	fork	0	0	stopped	0%	0b	root	disabled

```
[root@ip-172-31-30-133 sample_nodejs_on_Ec2]# pm2 start index.js --name node-app
```

```
[PM2] Applying action restartProcessId on app [node-app](ids: [ 0 ])
```

```
[PM2] [node-app](0) ✓
```

```
[PM2] Process successfully started
```

id	name	namespace	version	mode	pid	uptime	status	cpu	mem	user	watching
0	node-app	default	1.0.0	fork	27149	0s	online	0%	6.2mb	root	disabled

id	name	namespace	version	mode	pid	uptime	status	cpu	mem	user	watching
0	node-app	default	1.0.0	fork	27149	0s	online	0%	6.2mb	root	disabled

console picture :

The screenshot shows the AWS CloudShell interface with the following content:

```
[root@ip-172-31-30-133 sample_nodejs_on_Ec2]# ^C
[root@ip-172-31-30-133 sample_nodejs_on_Ec2]# pm2 stop index.js --name node-app
[PM2] Applying action stopProcessId on app [index.js](ids: [ 0 ])
[PM2] [node-app](0) ✓

id name namespace version mode pid uptime status cpu mem user watching
0 node-app default 1.0.0 fork 0 0 0 stopped 0% 0b root disabled

[root@ip-172-31-30-133 sample_nodejs_on_Ec2]# pm2 status node-app

id name namespace version mode pid uptime status cpu mem user watching
0 node-app default 1.0.0 fork 0 0 0 stopped 0% 0b root disabled

[root@ip-172-31-30-133 sample_nodejs_on_Ec2]# pm2 start index.js --name node-app
[PM2] Applying action restartProcessId on app [node-app](ids: [ 0 ])
[PM2] [node-app](0) ✓
[PM2] Process successfully started

id name namespace version mode pid uptime status cpu mem user watching
0 node-app default 1.0.0 fork 27149 0s 0 online 0% 6.2mb root disabled

[root@ip-172-31-30-133 sample_nodejs_on_Ec2]#
```

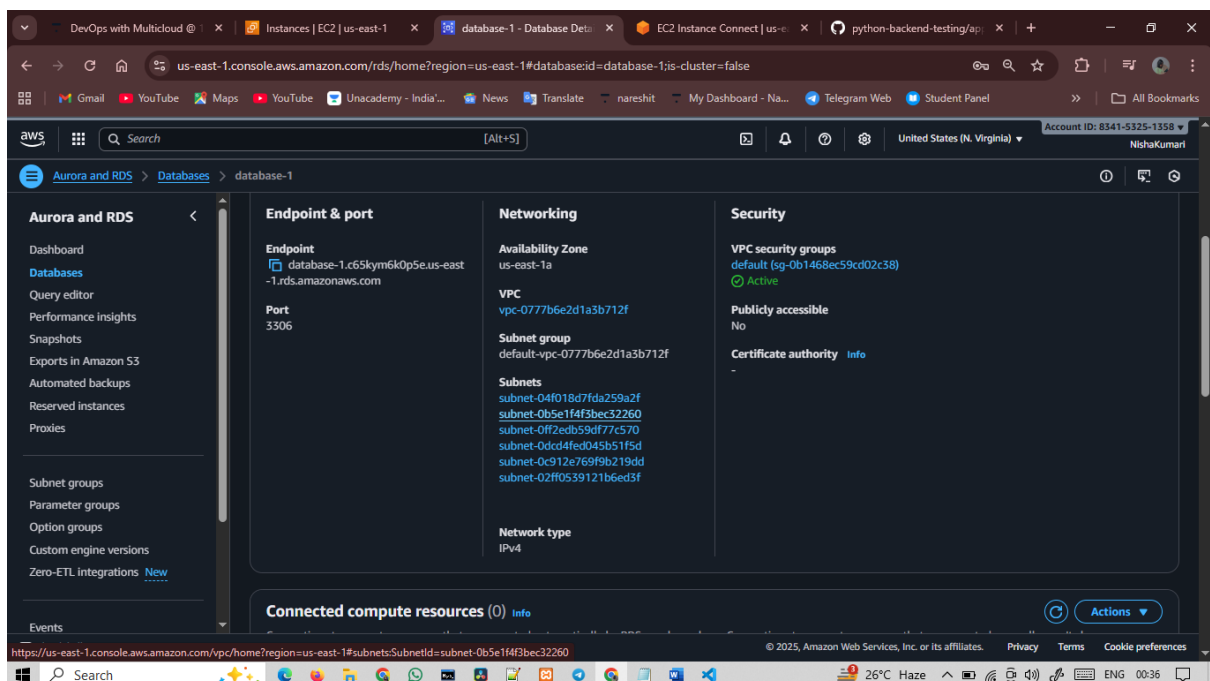
Below the terminal output, a summary box for the application is displayed:

```
i-003f13bdb22fac9e0 (nodejs)
PublicIPs: 54.91.61.15 PrivateIPs: 172.31.30.133
```

The bottom of the screenshot shows the Windows taskbar with the date and time as 23:40 on 26°C Haze.

Backend communication with database:--

create a database :--



,#_

~_####_ Amazon Linux 2023

~~_#####\

~~_###|

~~_#/___

<https://aws.amazon.com/linux/amazon-linux-2023>

~~_V~' '->

~~~/

~~\_.\_.\_/

\_\_/\_/\_/

\_\_/\_/m/'

Last login: Mon Sep 22 17:31:55 2025 from  
18.206.107.27

[ec2-user@ip-172-31-30-133 ~]\$ sudo su

[root@ip-172-31-30-133 ec2-user]# sudo su-

sudo: su-: command not found

[root@ip-172-31-30-133 ec2-user]# sudo su -

Last login: Mon Sep 22 18:58:58 UTC 2025 on  
pts/3

[root@ip-172-31-30-133 ~]# yum install  
python3-pip -y

Last metadata expiration check: 1:26:17 ago on  
Mon Sep 22 17:34:16 2025.

Package python3-pip-21.3.1-  
2.amzn2023.0.13.noarch is already installed.

Dependencies resolved.

Nothing to do.

Complete!